



ALGEBRA 2

CLASS 13

Rational Expressions

Simplifying, Multiplying, and Dividing


$$\frac{P(x)}{Q(x)}$$
$$Q(x) \neq 0$$

Build confidence. Recognize patterns. Think mathematically.

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Understanding Rational Expressions

A **rational expression** is a fraction whose numerator and denominator are polynomials.

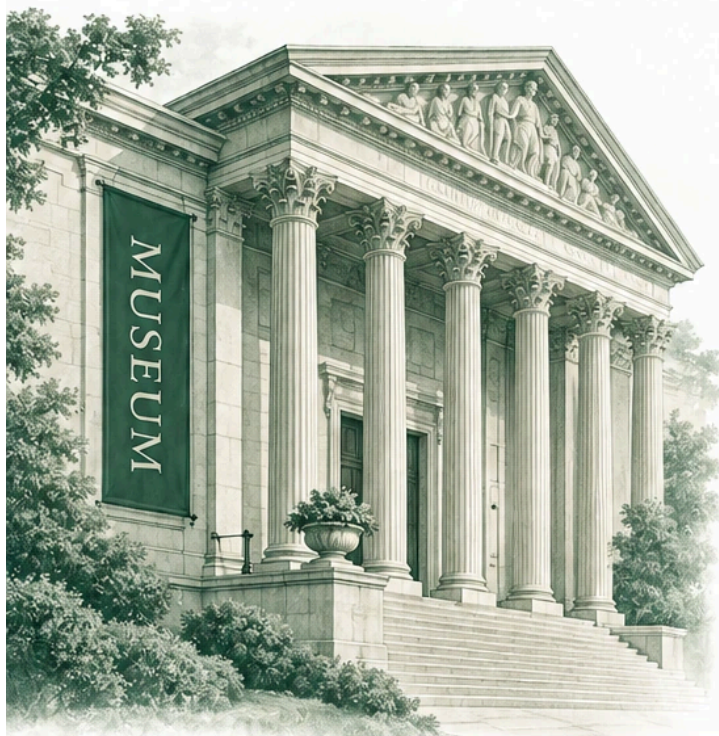
$$\frac{x + 2}{x - 3}$$

Numerator:
polynomial

Denominator:
polynomial

The denominator cannot equal zero.

Here, $x \neq 3$.



Real-world connection: A museum visit

Suppose a group pays a \$120 guide fee plus \$5 per student.

x = number of students.

Average cost per student =

$$\frac{120 + 5x}{x}$$

For 20 students:

$$\frac{120 + 5 \cdot 20}{20} = \$11 \text{ per student.}$$

In this context, x is a positive whole number.



MATH HINT

The numerator gives the total cost. Dividing by x gives the cost per student.

Image 1 practice: Understanding rational expressions

Q1. For $F(x) = \frac{5x - 2}{2x + 6}$, identify the numerator and denominator, find the excluded value, and evaluate $F(-2)$.

Q2. Evaluate $\frac{3x^2 - 5}{2x - 1}$ at $x = -2$ and $x = \frac{1}{2}$. Explain any undefined result.

Q3. For $\frac{x^2 - 9}{x^2 - x - 6}$, classify the result at $x = -3, 2, 3$ as zero, nonzero, or undefined. Give each defined value.

Q4. Which are quotients of polynomials: $\frac{x^2 + 2}{3x - 1}$, $\frac{\sqrt{x} + 2}{x - 1}$, and $\frac{x + 2}{\sqrt{9}}$? For each one that qualifies, state its real domain.

Q5. A workshop charges \$108 plus \$8 per student. A sponsor pays \$24 of the bill. Write the average amount paid by each of n students and compare the averages for 7 and 12 students.

Finding Excluded Values

Excluded values make the original denominator equal to zero. They are not allowed.

$$\frac{x + 1}{x^2 - 9}$$

1 Set the denominator equal to zero.

$$x^2 - 9 = 0$$

2 Factor.

$$(x - 3)(x + 3) = 0$$

3 Solve and exclude.

$$x = 3 \text{ or } x = -3$$

Restrictions: $x \neq 3$ and $x \neq -3$.

Domain: all real numbers except -3 and 3 .



Real-world connection: Cycling speed

A cyclist travels 12 miles in t hours.

$$\text{Average speed} = \frac{12}{t} \text{ mph.}$$

Here, $t \neq 0$.

Dividing by zero is undefined.

For this trip, elapsed time must be positive: $t > 0$.

If $t = 1.5$, the average speed is 8 mph.



MATH HINT

Check the **ORIGINAL** denominator before simplifying.
A canceled factor does not remove an excluded value.

Image 2 practice: Finding excluded values

Q6. Find all excluded values of $\frac{x+1}{3x^2-12x}$. Explain why both factors of the denominator matter.

Q7. Find the real domain of $\frac{x-2}{(2x-3)(x+5)}$, and decide whether $x = \frac{3}{2}$ is allowed.

Q8. Find every excluded value of $\frac{x^2-16}{x^2+2x-8}$. Which exclusion comes from a factor that cancels?

Q9. Find the real domain of $\frac{2x-1}{2x^2-7x+3}$. Is $x = \frac{1}{2}$ a zero of the original expression?

Q10. Find every excluded value of $\frac{x^2-9}{2x^3-18x}$, then simplify. Which restrictions would be missed by looking only at the simplified denominator?

Factoring Before Simplifying

Factor the numerator and denominator completely to reveal common factors.

$$\frac{x^2 + 5x + 6}{x^2 - 4}$$

1 Factor the numerator.

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

$2 + 3 = 5$ and $2 \times 3 = 6$.

2 Factor the denominator.

$$x^2 - 4 = (x + 2)(x - 2)$$

Use the difference of squares.

3 Rewrite using factors.

$$\frac{(x + 2)(x + 3)}{(x + 2)(x - 2)}$$

Common factor: $x + 2$.

Original restrictions: $x \neq -2$ and $x \neq 2$.



Real-world connection: Garden design

A rectangular garden has area $x^2 + 5x + 6$ square meters and width $x + 2$ meters, where $x > 0$.

$$\text{Length} = \frac{x^2 + 5x + 6}{x + 2} = x + 3 \text{ meters.}$$

Factoring reveals the missing dimension.



MATH HINT

Look for a greatest common factor first, then check for trinomials and special patterns.

Image 3 practice: Factoring before simplifying

Q11. Simplify and state every excluded value: $\frac{8x^2 + 24x}{4x}$.

Q12. Simplify and state every excluded value: $\frac{2x^2 - 50}{4x + 20}$.

Q13. Simplify and state every excluded value: $\frac{2x^2 + 11x + 12}{2x + 3}$.

Q14. Simplify and state every excluded value: $\frac{x^2 + 2x - 15}{x^2 - 6x + 9}$.

Q15. Simplify and state every excluded value: $\frac{6x^2 + 3x - 9}{4x^2 + 8x - 12}$.

Simplifying Rational Expressions

Cancel common factors, then keep every restriction from the original denominator.

$$\frac{(x+2)(x+3)}{(x+2)(x-2)}$$

1 Record the restrictions.

$$(x+2)(x-2) \neq 0$$

$$x \neq -2 \text{ and } x \neq 2.$$

2 Cancel the common factor.

$$\frac{\cancel{(x+2)}(x+3)}{\cancel{(x+2)}(x-2)}$$

$$\frac{x+3}{x-2} = 1 \text{ because } x \neq -2.$$

3 Write the simplified expression.

$$\frac{x+3}{x-2} \quad x \neq -2, 2$$

The expression changes form; the original restrictions remain.



Real-world connection: Printing posters

Suppose the total cost of printing n posters is $4n$ dollars, plus a setup fee of 20 dollars, where n is a positive whole number.

$$\text{Average cost} = \frac{4n + 20}{n} = 4 + \frac{20}{n}$$

Only the common factor in $4n/n$ cancels.
The setup fee is shared across the order.



MATH HINT

Cancel factors, not terms. Never cancel across addition or subtraction.

Image 4 practice: Simplifying and keeping restrictions

Q16. Simplify and state every excluded value: $\frac{18x^3}{24x^2}$.

Q17. Simplify and state every excluded value: $\frac{(x-4)^2(x+6)}{(x-4)(x+6)^2}$.

Q18. Simplify and state every excluded value: $\frac{4x^2 - 49}{2x^2 + 9x + 7}$.

Q19. Simplify and state every excluded value: $\frac{2x^2 - 12x + 18}{3x^2 - 27}$.

Q20. Simplify and state every excluded value: $\frac{6x^3 - 24x^2}{3x^3 - 24x^2 + 48x}$.

Avoiding Common Cancellation Mistakes

Cancel common factors, never individual terms joined by addition or subtraction.

✗ INCORRECT

$$\frac{x + 3}{x} \neq 3$$

The x in $x + 3$ is a term, not a factor of the whole numerator.

✓ CORRECT

$$\frac{3x}{x} = 3, x \neq 0$$

The numerator is a product: $3 \times x$. The common factor x cancels.

1 Factor first when possible.

$$\frac{3x + 6}{3} = \frac{3(x + 2)}{3} = x + 2$$

The factor 3 multiplies the entire numerator.

2 Check with a number.

At $x = 2$: $\frac{x + 3}{x} = \frac{5}{2}$, not 3.



Real-world connection: Movie tickets

Suppose n movie tickets cost \$12 each, plus one \$6 booking fee per order.

$$\text{Average cost per ticket} = \frac{12n + 6}{n} = 12 + \frac{6}{n}$$

$$\text{For 3 tickets: } 12 + \frac{6}{3} = \$14 \text{ per ticket.}$$

Here, n is a positive whole number.



MATH HINT

Ask: Does this factor multiply the ENTIRE numerator and denominator?

Image 5 practice: Avoiding cancellation mistakes

Q21. A student claims $\frac{3x + 12}{3x} = 4$. Disprove the claim at $x = 2$, then give a correct expression and restriction.

Q22. A student simplifies $\frac{6x^2 + 18x}{3x}$ to $2x + 18$. Correct the result and explain both cancellations.

Q23. Rewrite $\frac{6x^2 - 9x + 12}{3x}$ as separate terms. State its restriction and evaluate it at $x = -2$.

Q24. A student says $\frac{x^2 - 4x + 4}{x^2 - 4} = \frac{x - 2}{x + 2}$ for every $x \neq -2$. Correct the domain and compare the two formulas at $x = 2$.

Q25. A student cancels $4 - x$ with $x - 4$ in $\frac{(4 - x)(x + 3)}{x^2 - 16}$ and gets $\frac{x + 3}{x + 4}$. Find and correct the sign error; state all exclusions.

Multiplying Rational Expressions

Factor completely, cancel common factors, then multiply what remains.

$$\frac{x^2 - 4}{x + 1} \cdot \frac{x + 1}{x + 2}$$

General rule:

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

$$b \neq 0, d \neq 0$$

1

Record the restrictions.

Original denominators: $x + 1$ and $x + 2$
 $x \neq -1$ and $x \neq -2$.

2

Factor and cancel.

$$\frac{(x - 2)(x + 2)}{x + 1} \cdot \frac{x + 1}{x + 2}$$

The common factors $x + 1$ and $x + 2$ cancel.

3

Multiply the remaining factors.

$$x - 2$$

Restrictions remain: $x \neq -1, -2$.



Real-world connection: Travel time

A train travels d kilometers at v kilometers per hour.

Time = $\frac{d}{v}$ hours Distance = speed $\times$ time

$$v \cdot \frac{d}{v} = d$$

The factor v cancels. In this context, $d > 0$ and $v > 0$.



MATH HINT

Cancel common factors before multiplying to keep the algebra simpler.

Image 6 practice: Multiplying rational expressions

Q26. Simplify and state every excluded value: $\frac{6x^2}{5} \cdot \frac{15}{9x}$.

Q27. Simplify and state every excluded value: $\frac{2x+8}{x-1} \cdot \frac{x^2-1}{4x+16}$.

Q28. Simplify and state every excluded value: $\frac{x^2-49}{2x+6} \cdot \frac{3x+9}{x^2+9x+14}$.

Q29. Simplify and state every excluded value: $\frac{2x^2+7x+3}{x^2-9} \cdot \frac{x-3}{4x+2}$.

Q30. Simplify and state every excluded value: $\frac{3x^2-12}{2x^2+8x+8} \cdot \frac{x^2+5x+6}{x^2-4x+4}$.

Dividing Rational Expressions

Multiply by the reciprocal of the divisor. The divisor must be defined and cannot equal zero.

$$\frac{x^2 - 9}{x + 2} \div \frac{x - 3}{x + 2}$$

General rule:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

$$b \neq 0, c \neq 0, d \neq 0$$

1 Find all restrictions.

Denominators: $x + 2 \neq 0 \rightarrow x \neq -2$.

Divisor cannot be zero: $x - 3 \neq 0 \rightarrow x \neq 3$.

2 Multiply by the reciprocal.

$$\frac{x^2 - 9}{x + 2} \cdot \frac{x + 2}{x - 3}$$

Flip only the second fraction.

3 Factor, cancel, and simplify.

$$\frac{(x - 3)(x + 3)}{x + 2} \cdot \frac{x + 2}{x - 3} = x + 3$$

$x + 3$, with $x \neq -2$ and $x \neq 3$.



Real-world connection: Filling bottles

You have 6 liters of water. Each bottle holds $\frac{3}{4}$ liter.

$$\text{Number of bottles} = 6 \div \frac{3}{4} = 6 \times \frac{4}{3} = 8$$

Dividing by a fraction tells us how many equal portions fit.



MATH HINT

Check both original denominators and the numerator of the divisor before simplifying.

Image 7 practice: Dividing rational expressions

Q31. Simplify and state every excluded value: $\frac{5x^2}{12} \div \frac{10x}{9}$.

Q32. Simplify and state every excluded value: $\frac{2x + 10}{x - 3} \div \frac{x + 5}{x^2 - 9}$.

Q33. Simplify and state every excluded value: $\frac{x^2 - 64}{2x + 6} \div \frac{x - 8}{x^2 + 6x + 9}$.

Q34. Simplify and state every excluded value: $\frac{2x^2 + 9x + 10}{x^2 - 4} \div \frac{2x + 5}{x^2 - 2x}$.

Q35. Simplify and state every excluded value: $\frac{3x^2 - 27}{x^2 + 4x + 3} \div \frac{2x - 6}{x^2 + 5x + 4}$.

Combining Multiplication and Division

Rewrite division as multiplication by a reciprocal, factor, and simplify.

$$\frac{x^2 - 1}{x} \cdot \frac{x}{x + 2} \div \frac{x - 1}{x + 2}$$

1 Record every restriction.

Original denominators: $x \neq 0$ and $x \neq -2$.
The divisor cannot be zero: $x \neq 1$.

2 Rewrite division and factor.

$$\frac{(x - 1)(x + 1)}{x} \cdot \frac{x}{x + 2} \cdot \frac{x + 2}{x - 1}$$

Take the reciprocal of the divisor only.

3 Cancel common factors.

Cancel x , $x + 2$, and $x - 1$.

$$x + 1$$

Restrictions remain: $x \neq 0, -2, 1$.

Check at $x = 3$: $\frac{8}{3} \cdot \frac{3}{5} \div \frac{2}{5} = 4$.



Real-world connection: Scaling a recipe

One batch uses $\frac{3}{4}$ cup of flour and makes 6 servings. How much flour is needed for 10 servings?

$$\frac{3}{4} \times 10 \div 6 = \frac{3}{4} \times \frac{10}{6} = \frac{5}{4} \text{ cups}$$

That is $1\frac{1}{4}$ cups of flour.



MATH HINT

Multiplication and division have equal priority. Work left to right, or rewrite division using reciprocals first.

Image 8 practice: Combining multiplication and division

Q36. Simplify and state every excluded value: $\frac{4x^2}{9} \cdot \frac{15}{2x} \div \frac{5x}{6}$.

Q37. Simplify and state every excluded value: $\frac{2x+6}{x-4} \cdot \frac{x^2-16}{x+1} \div \frac{x+3}{2x+2}$.

Q38. Simplify and state every excluded value: $\frac{x^2-49}{2x} \cdot \frac{3x}{x+2} \div \frac{x-7}{2x+4}$.

Q39. Simplify and state every excluded value: $\frac{2x^2+11x+12}{x^2-16} \cdot \frac{x-4}{x+6} \div \frac{2x+3}{3x+18}$.

Q40. Simplify and state every excluded value: $\frac{x^2-36}{x^2-4} \div \frac{2x-12}{x+2} \cdot \frac{x^2-4x+4}{x+5}$.

Homework review: R1–R5

R1. For $G(x) = \frac{2x^2 - 3x - 2}{2x + 1}$, find the excluded value and evaluate $G(-1)$ and $G(2)$.

R2. Find all exclusions of $\frac{x - 3}{x(2x^2 - 5x - 3)}$. Would canceling a common factor ever restore an excluded input?

R3. Simplify and state every excluded value: $\frac{3x^2 + 18x + 24}{2x^2 + 4x}$.

R4. Simplify and state every excluded value: $\frac{4x^2 - 36}{2x^2 - 12x + 18}$.

R5. A student claims $\frac{5 - x}{x^2 - 25} = \frac{1}{x + 5}$. Correct the sign and restrictions, then evaluate at $x = 0$.

Homework review: R6–R8

R6. Simplify and state every excluded value: $\frac{2x^2 - 50}{x^2 + 6x + 5} \cdot \frac{x + 1}{4x - 20}$.

R7. Simplify and state every excluded value: $\frac{x^2 + 9x + 20}{2x^2 - 32} \div \frac{x + 5}{x^2 - 3x - 4}$.

R8. Simplify and state every excluded value: $\frac{3x^2 - 48}{x^2 - 1} \cdot \frac{x - 1}{2x + 6} \div \frac{x - 4}{x^2 + 4x + 3}$.

Homework word problems: W1–W5

W1. A pottery club pays \$120 plus \$9 per participant, then receives a \$30 group discount. Write the average cost for n participants. How much less does each person pay with 18 participants than with 12?

W2. A cyclist rides 24 miles to a park in t hours and returns along the same route in $t + \frac{1}{2}$ hours. Write the average speed for the entire trip and evaluate it when $t = \frac{3}{2}$. State a practical domain.

W3. A rectangular mat has area $2x^2 + 13x + 15$ square meters and width $2x + 3$ meters, with $x > 0$. Find its length in simplest form and its perimeter when $x = 4$.

W4. A display has remaining area $4x^2 - 25$ square meters after a panel is removed. It is rearranged into a rectangle of width $2x - 5$ meters, with $x > \frac{5}{2}$. Find its length and a simplified formula for its perimeter.

W5. A print shop charges $12n + 60$ dollars for n posters and applies a \$24 order discount. A student says the average cost is always \$48. Correct the claim and compare the averages for 9 and 18 posters.

Homework word problems: W6–W10

W6. A sign has length $\frac{x^2 + 7x + 10}{x^2 - 1}$ meters and width $\frac{3(x - 1)}{x + 2}$ meters, where $x > 1$.

Simplify its area, state all algebraic exclusions, and find the area at $x = 3$.

W7. A machine produces $\frac{6x}{x + 4}$ parts per minute for $\frac{x^2 + 7x + 12}{2x}$ minutes, where x is a positive whole number. Simplify the number produced. At $x = 5$, how many usable parts remain if three are rejected?

W8. A ribbon is $\frac{2x^2 - 72}{x + 2}$ meters long. Each piece is $\frac{x - 6}{x + 2}$ meter long, where x is a whole number greater than 6. Find the number of pieces, then the number available at $x = 8$ after four pieces are reserved.

W9. A tank contains $\frac{2x^2 + 11x + 12}{x + 1}$ liters. Each container holds $\frac{2x + 3}{2x + 2}$ liters, with x a positive whole number. How many containers can be filled? At $x = 2$, how many are left for sale after one is kept as a sample?

W10. A panel has area $\frac{x^2 - 16}{x + 5}$ square meters and width $\frac{2x - 8}{x + 2}$ meters, with $x > 4$. Simplify its length and evaluate at $x = 6$. Explain why $x = 4$ is forbidden even if it disappears during cancellation.

Homework word problems: W11–W12

W11. A recipe uses $\frac{x^2 + 8x + 15}{x + 1}$ cups of flour for $2(x + 3)$ servings, with x a positive whole number. At the same rate, simplify the flour needed for $3(x + 1)$ servings and evaluate at $x = 3$.

W12. A machine produces $\frac{2x^2 - 50}{x + 1}$ meters of ribbon per minute for $\frac{x^2 + 4x + 3}{3(x + 2)}$ minutes. Each spool holds $\frac{x^2 - 25}{x + 2}$ meters, with $x > 5$. Simplify the number of spool-capacities produced. At $x = 7$, how many full spools and what fraction of another can be filled?